

Module: Window server security and maintenance

1. Explain the process of installing and configuring Hyper-V virtualization in Windows Server 2016

Ans-What is Hyper-V?

Hyper-V is Microsoft's virtualization platform used to create and manage virtual machines (VMs).

Installation Steps

1. Open Server Manager
2. Click Add Roles and Features
3. Select Role-based installation
4. Choose your server
5. Select Hyper-V
6. Click Add Features
7. Next → Select virtual switch
8. Finish installation
9. Restart server

Configuration

After restart:

Open Hyper-V Manager

Create VM:

1. Click New → Virtual Machine
2. Enter VM name
3. Select Generation
4. Assign RAM
5. Configure network
6. Create virtual hard disk
7. Select ISO file
8. Finish

Start VM and install OS.

2. How do you monitor server performance and manage event logs in Windows Server?

Ans-Performance Monitoring

Use Performance Monitor

Open:

perfmon

Monitor:

- CPU usage
- RAM usage
- Disk activity
- Network performance

Useful tools:

- Task Manager
- Resource Monitor
- Performance Monitor

Manage Event Logs

Open:

eventvwr

Logs include:

- Application
- Security
- System
- Setup

Used for:

- Troubleshooting errors
- Security monitoring
- Tracking system events

3. Describe the different types of storage options available in Windows Server

Ans- 1. DAS (Direct Attached Storage)

Storage directly attached to server.

Example:

- HDD
- SSD

Fast but limited sharing.

2. NAS (Network Attached Storage)

Storage connected through network.

Used for:

- File sharing
- Backup

3. SAN (Storage Area Network)

High-speed dedicated storage network.

Used in enterprises.

Benefits:

- High performance
- Centralized storage

4. Cloud Storage

Online storage.

Examples:

- Microsoft Azure
- OneDrive

4. What is the role of File Server and how do you configure it?

Ans- Role

File Server stores and shares files across network.

Benefits:

- Centralized storage
- Access control
- Easy backup

Configuration Steps

1. Server Manager
2. Add Roles → File Server
3. Install

Create shared folder:

1. Create folder:

D:\SharedData

2. Right click → Properties → Sharing
3. Advanced Sharing
4. Check Share this folder
5. Set permissions

Access from client:

\\ServerName\SharedData

5. Explain DFS in Windows Server 2016

Ans-DFS Full Form

Distributed File System

DFS allows multiple shared folders to appear as one shared folder.

Types

- DFS Namespace
- DFS Replication

Configuration

1. Install DFS Namespace
2. Open DFS Management

3. Create Namespace
4. Add shared folders
5. Configure replication

Benefits:

- Easy access
- Fault tolerance
- Data replication

6. Discuss built-in backup and recovery options

Ans-Windows Server provides:

1. Windows Server Backup

Full backup tool.

2. System State Backup

Backs up:

- Registry
- Active Directory
- Boot files

3. Bare Metal Recovery

Restores full server.

4. File Recovery

Restore deleted files.

7. How do you configure Windows Server Backup?

Ans- Install Feature

Server Manager → Add Features →
Windows Server Backup

Configure Backup

1. Open Windows Server Backup
2. Select Backup Schedule
3. Choose:

- Full Server
 - Custom Backup
4. Select destination
 5. Schedule time
 6. Finish

8. Explain restoring files using Windows Server Backup

Ans- Steps:

1. Open Windows Server Backup
2. Click Recover
3. Select backup location
4. Select backup date
5. Choose:
 - Files/Folders
 - Volumes
 - Applications
6. Select restore destination
7. Start restore

9. Common troubleshooting techniques for Windows Server startup issues

Ans- Check Hardware

- RAM
- HDD
- Power supply

Safe Mode

Press:

F8

Choose:

- Safe Mode
- Last Known Good Configuration

Startup Repair

Boot from installation media

Select:

Repair your computer

Check Event Viewer

eventvwr

Run Repair Commands

sfc /scannow

chkdsk /f /r

bootrec /fixmbr

bootrec /fixboot

10. How do you troubleshoot network connectivity problems?

Ans- Check IP

ipconfig

Test Connectivity

ping 127.0.0.1

ping gateway-ip

ping google.com

Check DNS

nslookup google.com

Release/Renew

ipconfig /release

ipconfig /renew

Reset Network

netsh int ip reset

11. Common Active Directory issues and troubleshooting

Ans- Problem: User cannot login

Check:

- Username/password
- Account locked?

Unlock in:

Active Directory Users and Computers

Replication Issue

Check:

repadmin /replsummary

DNS Issue

Check:

dcdiag

nslookup

Service Issue

Restart:

net stop ntds

net start ntds

12. Troubleshoot performance problems on Windows Server

Ans:Check CPU / RAM

Open:

taskmgr

Resource Monitor

resmon

Performance Monitor

perfmon

Check Services

Stop unnecessary services:

services.msc

Disk Cleanup

Delete temp files:

cleanmgr